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BS-CYBER SECURITY-FALL-24

SUBJECT:-

PROGRAMMING FUNDAMENTALS (Lab)

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PROVIDED TO

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LAB #4**Lab Task 1: Understanding Basic Data Types****Code:**

```
> [*] L4 T1.cpp
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      //Declaring different data types with variables
6      int i;
7      float f;
8      double d;
9      char c;
10     bool b;
11
12     //Taking values from user
13     cout<<"Enter an Integral value " <<endl;
14     cin >> i;
15     cout << "Enter a Floating Value " <<endl;
16     cin >> f;
17     cout << "Enter a double value " <<endl;
18     cin >> d;
19     cout << "Enter a Character " <<endl;
20     cin >> c;
21     cout << "Enter a bool value " <<endl;
22     cin >> b;
23
24     //printing out the values
25     cout << endl << "Integer is " << i <<endl ;
26     cout << "Float is " << f << endl ;
27     cout << "Double is " << d << endl;
28     cout << "Character is " << c << endl;
29     cout << "Bool is " << b << endl;
30
31     return 0;
32 }
```

Output:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 4\L4 T1.exe
Enter an Integral value
5
Enter a Floating Value
7.987
Enter a double value
9.07564343
Enter a Character
A
Enter a bool value
9

Integer is 5
Float is 7.987
Double is 9.07564
Character is A
Bool is 1

-----
Process exited after 39.69 seconds with return value 0
Press any key to continue . . .
```

Lab Task 2: Type Conversion (Implicit and Explicit)

Code:

```
[*] L4 T1.cpp  x  L4 T2.cpp  x
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      //Declaring and initializing variables
6      int a = 10 ;
7      float b ;
8
9      //assigning value
10     b = a ;
11
12     //Performing conversion
13     cout << "Implicit type Conversion (Int to Float) is " << b << endl ;
14
15     //Declaring and initializing variables
16     int c ;
17     float d = 7.678 ;
18
19     //assigning value
20     c = d ;
21
22     //Performing Conversion
23     cout << "Explicit type Conversion (Float to Int) is " << c << endl ;
24
25     return 0 ;
26 }
```

Output:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 4\L4 T2.exe
Implicit type Conversion (Int to Float) is 10
Explicit type Conversion (Float to Int) is 7
-----
Process exited after 0.1215 seconds with return value 0
Press any key to continue . . .
```

Lab Task 3: Character to ACSII and Vice Versa,**Code:**

```
[*] L4 T1.cpp  ×  L4 T2.cpp  ×  L4 T3.cpp  ×
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      //Declaring Variables
6      char c, d;
7      int i;
8
9      //Taking input from user
10     cout << "Enter Any Character " << endl ;
11     cin >> c ;
12
13     //Printing ASCII code
14     cout << "ASCII value of character is "
15     << int ( c ) << endl ;
16
17     //Converting ASCII code to Character
18     cout << "\n Enter ASCII code of any Character " << endl ;
19     cin >> i ;
20     d = i ;
21     cout << "The Character of entered ASCII value is " << d << endl ;
22
23     return 0 ;
24 }
```

Output:

```
L4 T1.cpp  ×  L4 T2.cpp  ×  L4 T3.cpp  ×
1  D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 4\L4 T3.exe
2  Enter Any Character
3  A
4  ASCII value of character is 65
5
6  Enter ASCII code of any Character
7  97
8  The Character of entered ASCII value is a
9
10 -----
11 Process exited after 7.917 seconds with return value 0
12 Press any key to continue . . .
13
14
```

Lab Task 4: Data Type Size Checker

Code:

```
[*] L4 T1.cpp  ×  L4 T2.cpp  ×  L4 T3.cpp  ×  L4 T4.cpp  ×
1  # include <iostream>
2  using namespace std;
3  int main ()
4  {
5      cout << "The Size of Integer Data Type is "
6      << sizeof ( int ) << " Bytes " << endl ;
7
8      cout << "The Size of Float Data Type is "
9      << sizeof ( float ) << " Bytes " << endl ;
10
11     cout << "The Size Double Data Type is "
12     << sizeof ( double ) << " Bytes " << endl ;
13
14     cout << "The size of Char Data Type is "
15     << sizeof ( char ) << " Bytes " << endl ;
16
17     cout << "The size of Bool Data Type is "
18     << sizeof ( bool ) << " Bytes " << endl ;
19
20     return 0 ;
21 }
```

Output:

```
Search View Project Execute Tools AStyle Window Help
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 4\L4 T4.exe
The Size of Integer Data Type is 4 Bytes
The Size of Float Data Type is 4 Bytes
The Size Double Data Type is 8 Bytes
The size of Char Data Type is 1 Bytes
The size of Bool Data Type is 1 Bytes

-----
Process exited after 0.1152 seconds with return value 0
Press any key to continue . . .
```

THANK YOU FOR YOUR TIME !

THE END